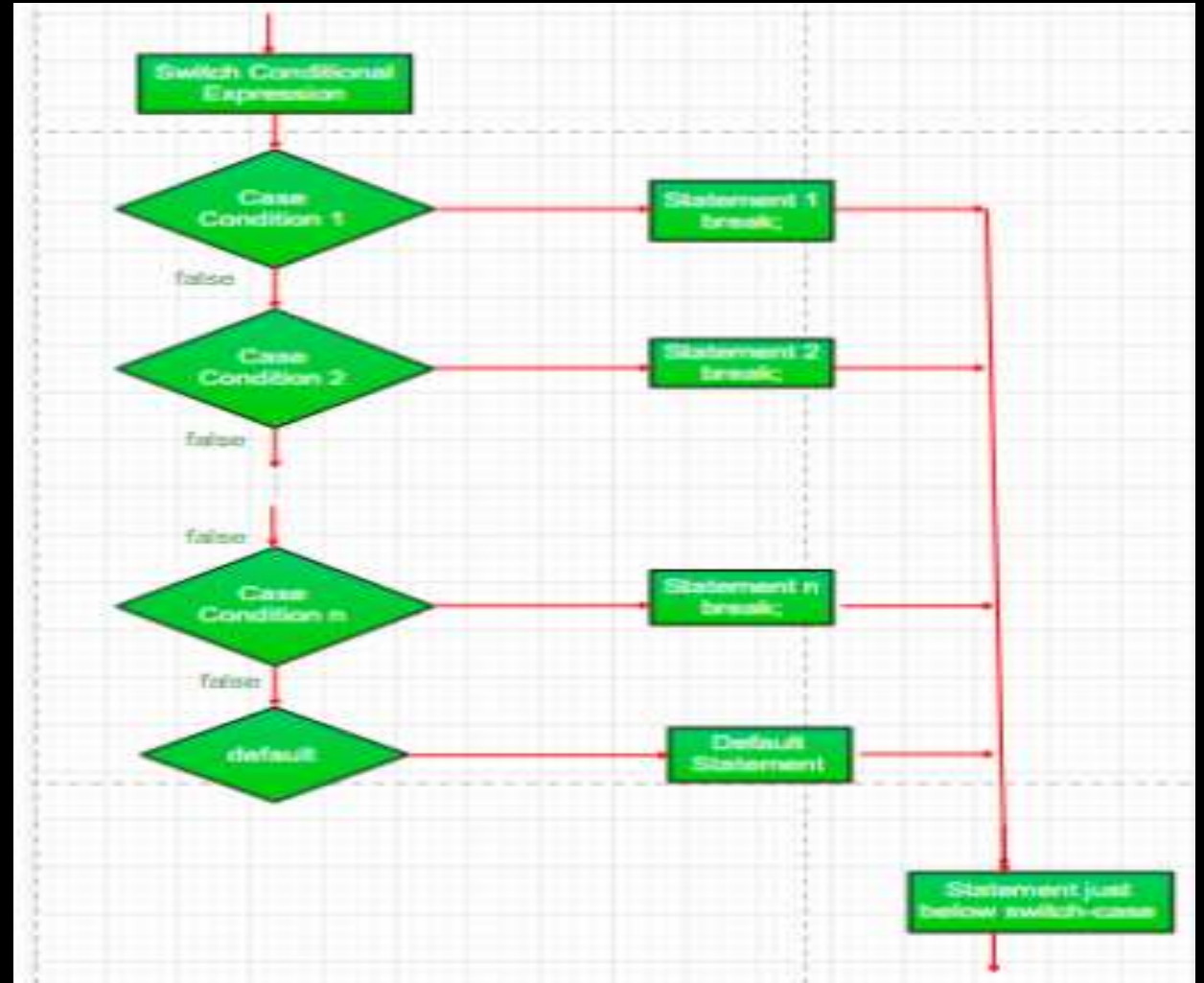


c
language

Control structure

Switch Case structure

- The switch statement is a multiway branch statement. It provides an easy way to dispatch execution to different parts of code based on the value of the expression.



Switch Case structure

- **Every time C language encounters a *switch* statement**

1. It evaluates the **switch expression**, and jumps to the case whose selector matches the value of the expression.
2. The program executes the statements in order from that point on until a break statement is encountered, skipping then to the first statement after the end of the switch structure.
3. If none of the cases are satisfied, the default block is executed. However, note that the default part is optional.

- Unlike with the if statement, the multiple statements are executed in the switch statement without needing the curly braces.
- When a case in a switch statement has been matched, all the statements associated with that case are executed. Not only that, the statements associated with the succeeding cases are also executed.
- To prevent the program from executing statements in the subsequent cases, use a break statement as the last statement.

Switch Case structure

- Expression can be of type byte, short, int char or an enumeration.
- Duplicate case values are not allowed.
- The default statement is optional.
- The break statement is used inside the switch to terminate a statement sequence.
- The break statement is optional. If omitted, execution will continue on into the next case.

```
#include <stdio.h>
int main()
{
    int i = 9;
    switch (i)
    {
        case 0:
            printf("i is zero.");
            break;
        case 1:
            printf("i is one.");
            break;
        case 2:
            printf("i is two.");
            break;
        default:
            printf("i is greater than 2.");
    }
}
```

Switch Case structure

```
int grade = 92;

switch (grade) {
    case 100:
        printf("Excellent!");
        break;
    case 90:
        printf("Good!");
        break;
    case 80:
        printf("Study Harder!");
        break;
    default:
        printf("You failed!");
}
```

```
#include<stdio.h>
int main()
{
    int i = 9;
    switch (i)
    {
        case 0:
            printf("i is zero.");
            break;
        case 1:
            printf("i is one.");
            break;
        case 2:
            printf("i is two.");
            break;
        default:
            printf("i is greater than 2.");
    }
}
```